

KANCHANA GAYATHRI ARIYASINGHE

SOFTWARE ENGINEER

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PROFILE

Final-year Computer Engineering undergraduate at the University of Jaffna with hands-on experience in full-stack development and AI based systems, including scalable web applications, machine learning, and computer vision techniques. Always adept at exploring new technologies and applying them to innovative solutions. Demonstrates a commitment to continuous learning and effective teamwork.

EDUCATION

University of Jaffna	Sri Lanka
B.Sc.Eng (Hons) in Computer Engineering (Undergraduate)	April 2022 (Present)
Cumulative GPA: 3.38 / 4.00	

SKILLS SUMMARY

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| <ul style="list-style-type: none">• Programming Languages: Java, C++, Python• Frameworks: React, Node.js, Spring Boot, Next.js, Strapi• Databases: MYSQL, MongoDB, PostgreSQL• Embedded Systems: Microcontroller Programming | <ul style="list-style-type: none">• Tools & Frameworks: Git, Jira, BitBucket• Problem Solving• Teamwork• Languages: Sinhala (Native), English |
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WORK EXPERIENCE

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|--|-------------------------------------|
| Software Engineer Intern – Sri Lanka Telecom | November 2025 – May 2026 (6 months) |
| Business Continuity Management System (BCMS) project - MERN stack | |
| <ul style="list-style-type: none">• Worked as a Full Stack Developer (Team Lead) on the BCMS project, leading the development team, managing the Bitbucket repository, and coordinating tasks. Contributed to full-stack feature development, and fixed critical bugs when integrating developments into hosted production server. | |
| Learning Path Management System (LPMS) project - PERN stack | |
| <ul style="list-style-type: none">• Worked as a Full Stack Developer on the LPMS for the HR Talent Development section. Contributed to ERP integration by fetching and managing API data for employee and learning related modules. Gained experience in API integration, database management, and full-stack development while collaborating with the team. | |

PROJECTS

- | | |
|--|--------------------------------------|
| Potholes Detection using Advanced Object Detection Techniques (Ongoing) | January 2025 (Present) |
| Researcher | |
| <ul style="list-style-type: none">• Developed a real-time pothole detection system using YOLOv8m and MobileNetV2, achieving 90.3% accuracy in low-light, rainy, and cluttered conditions. | |
| Hotel Management System | GitHub November 2025 |
| <ul style="list-style-type: none">• Developed a full-stack hotel management system with customer features for browsing hotel details, viewing gallery images, and making online room bookings using the MERN stack, Tailwind CSS, and JWT for secure authentication.• Implemented role-based functionalities for Receptionist (booking management), Housekeeping (room cleaning, maintenance, and status updates), and Admin (managing rooms, staff, and overall operations). | |

E-Commerce Web Application (Ongoing Project)

[GitHub](#) January 2026

- Developing a full-stack e-commerce web application for managing and purchasing computer products using the MERN stack, Tailwind CSS, and JWT for secure authentication.
- Implemented user functionalities including product search, add-to-cart, checkout, order placement, and profile management.
- Developing admin features for managing products, users, orders, and monitoring customer reviews.
- Planning to deploy the application on a free cloud hosting platform for public access and testing.

AI-Powered Video Learning Assistant (Ongoing Project)

[GitHub](#) April 2026

- Currently developing a full-stack AI based learning platform that transforms educational videos into structured study materials.
 - Implemented core features including video input handling, transcript processing, and AI generated summaries.
 - Currently enhancing the system with quiz generation, semantic search, and personalized learning tracking.
- Technologies: Next.js, Strapi, PostgreSQL, OpenAI API

Computer Lab Booking and Equipments Borrowing System

[GitHub](#) January 2025

Role: Project Manager and Developer, Software Engineering

- Developed a full-stack web application named LabMate for computer lab booking and equipment borrowing using the MERN stack, Vite, Tailwind CSS, and JWT for secure authentication. Led the project through Jira-based sprint management, designed the admin frontend with React.js, and contributed to backend development with Node.js and MongoDB for efficient, real-time data handling.

Research AI

[GitHub](#) November 2025

- Developed intelligent research assistant chatbot that analyzes PDFs, discovers relevant academic papers and datasets, and provides AI-powered answers using Groq, integrating arXiv, CrossRef, and Hugging Face APIs with a Flask backend and a JavaScript-based frontend.

Student Feedback Management System

[LinkedIn](#) January 2024

- Developed a Student Feedback Management System to efficiently collect and manage course and lecturer feedback within the faculty, featuring secure authentication for students, lecturers, and administrators, and built using PHP, HTML/CSS, and MySQL with phpMyAdmin for robust data management.

CERTIFICATION & COURSES

- Java Programming Internship (One month) – [Codesoft](#)
- AlgoXplore 1.0 Algorithm and Capture the Flag (CTF) Challenge - [NSBM](#)
- MoraXtreme 9.0, a 12-hour coding marathon - [IEEE UOM](#)
- TECHNOVATION 2025 | SLT-MOBITEL – AI Innovation Challenge (Finalist) – [SLT-MOBITEL](#)

VOLUNTEERING

- IESL Student Chapter - University of Jaffna
- IEEE Student Member

Membership Number: S-29763

Membership Number: 10041145

REFERENCES

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